

POLS 641

Introductory Political Analysis

Class 2: Basics of Causality

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Today's Agenda

- Causality
 - The importance of Causality
 - Experiments as an ideal
 - The core model
- Problem set # 1
- Final Project Introduction

Causality - Today's Focus

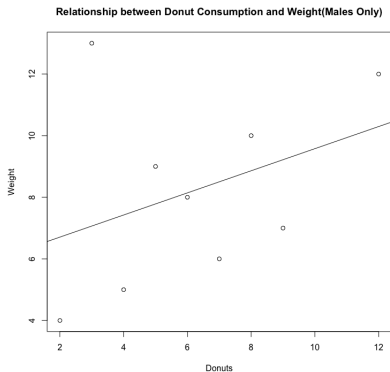
Why do we care?

Causality

Book Example: Donut consumption causes weight gain

Hypothesis: An increase in donut consumption will be associated with an increase in weight gain

Causality



Can we make a causal claim?

What would allow us to make a causal claim?

Why Causality?

Many (most?) interesting questions in political science are about *causes*!

- does democracy cause improvement in wealth, or health, or gender equality, or 'X'?
- do politicians change policies because of public opinion?
- does peacekeeping or development aid reduce violence?
- does education increase political participation?
- ...what does increase political participation?
- ...and many more questions.

A Classic Example

Hospitals:

How could we test the relationship between hospital stays and lifespan?

What is our IV and what is our DV?

- Independent variable or treatment: going to hospital
- Dependent variable or outcome: five year survival

What are we likely to find? We will likely see that going to the hospital will predict lower five year survival.

Does that mean going to the hospital *causes* lower survival?

No!

You should certainly not write a paper or give policy advice that hospitals are bad!

Three Ways To Ponder This

- ① **Selection problem:** people go to the hospital when they are sicker.
- ② **Confounding:** a confounder is something related to both the IV and the DV. *Sicker people* are both more likely to go to the hospital and have lower survival rates.
- ③ **Comparability (in absence of treatment).** Ask yourself: how might *those who go to the hospital* and *those who do not* differ on things other than the thing you are looking at (that they are in the hospital)?

Yet more ways to think about it!

You see a relationship between independent variable **A** and dependent variable, **B**

For example, you see countries that are **democracies** have higher average **income**.

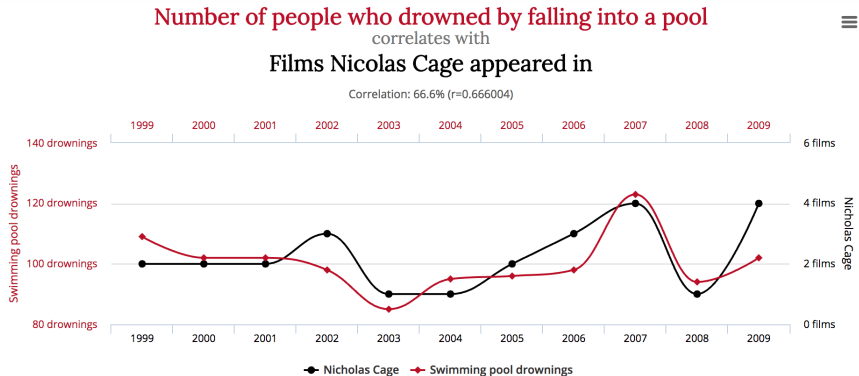
Maybe this is just due to chance – we will learn some significance tests to deal with that!

But if it is not due to chance, three possibilities remain:

- ① **A** (higher democracy) → **B** (higher income)
- ② **B** (higher income) → **A** (higher democracy)
- ③ **C** (unobserved confounder) → **A** & **B**!

In general, it is easy(ish) to tell if A and B are significantly related, but very difficult to tell the difference between these causal arrangements.

Correlation $\neq$ Causation

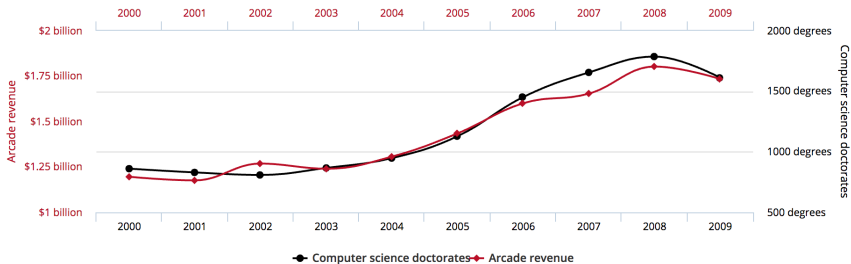


Data sources: Centers for Disease Control & Prevention and Internet Movie Database

Correlation $\neq$ Causation

Total revenue generated by arcades correlates with Computer science doctorates awarded in the US

Correlation: 98.51% ($r=0.985065$)



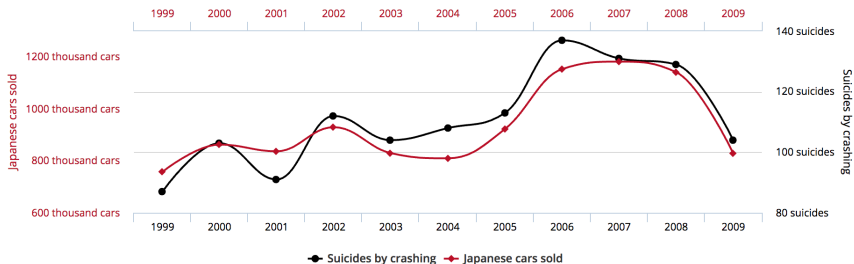
Data sources: U.S. Census Bureau and National Science Foundation

tylervgen.com

Correlation $\neq$ Causation

Japanese passenger cars sold in the US correlates with Suicides by crashing of motor vehicle

Correlation: 93.57% ($r=0.935701$)



Data sources: U.S. Bureau of Transportation Statistics and Centers for Disease Control & Prevention

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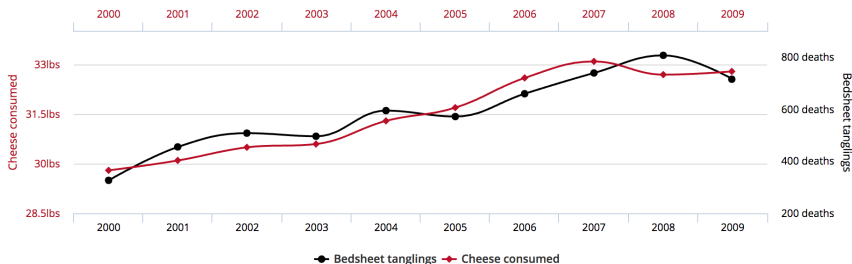
Correlation $\neq$ Causation

Per capita cheese consumption

correlates with

Number of people who died by becoming tangled in their bedsheets

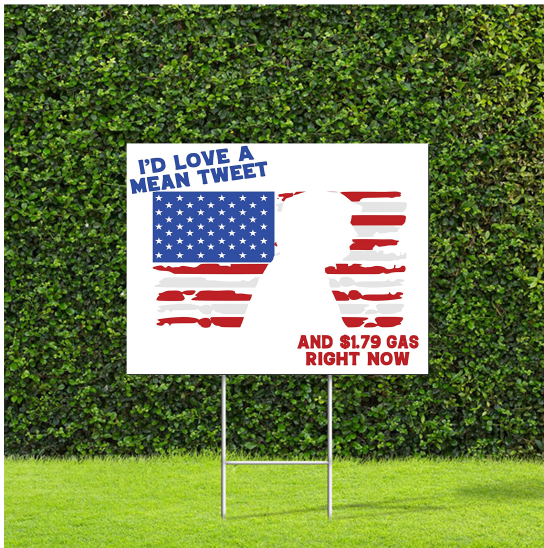
Correlation: 94.71% ($r=0.947091$)



Data sources: U.S. Department of Agriculture and Centers for Disease Control & Prevention

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Correlation $\neq$ Causation



Experiments

Example: In a survey you find that those who say they watched more political ads were more likely to politically participate. We also find it's a statistically significant relationship ($p < 0.001$, more on this later).

- ① A (seeing ads) $\rightarrow$ B (participation)
- ② B (participation) $\rightarrow$ A (seeing ads)
- ③ C (unobserved confounder) $\rightarrow$ A & B!

You *can* decide among these three if you—as the researcher—cause A or B.

Randomization: you randomly assign people to see ads (the IV):

- You could rule out reverse causality (#2)
- You can rule out any **confounders** (#3)
- You know there is no **selection effect** because you randomly assigned.
- In terms of **comparability**: ask yourself, in the absence of the ads, would the treated and control groups look any different? Especially on participation?

Randomness vs. Randomization

- The textbook uses both terms. It's a bit confusing.
- Important: These two terms do not mean the same thing.
 - Randomization (aka random assignment, RCT): using a coin toss to create treatment and control groups; creates exogeneity. ie we expect groups to be equal on all potential confounders(given adequate sample size)
 - Randomness: noise in the data (i.e. effects from sampling on a given day) - could go away with larger sample sizes. We will address some of these concerns with t-tests (p-values) and confidence intervals.
 - Random Variables: We will discuss next week. Variable with a known range and probabilities defined by a knowable distribution.

Terminology

Treated and Control:

- **Treated**: those we get some treatment of interest. e.g. those who get democracy, political ads, coffee.
- **Controls**: are those who do not get the “treatment” of interest. e.g. not on autocracy, no political ads, coffee.
- **Observational study**: general term for research where you don’t get to *randomize* who gets the treatment. Instead, you just observe some relationship in the world. **Check**: Typically, can you make causal claims from observational studies?
- **Experimental Study** and **Randomized Control Trial (RCT)**: common terms for research designs in which you do randomize who gets the treatment. **Check**: Typically, can you make causal claims from experimental studies?
- **Quasi-experimental research**: research in which you have observational data, but you find ways to ensure that the treatment was effectively (‘as-if’) randomly distributed.

Endogeneity Another way of saying correlation $\neq$ causation

- Endogeneity: The IV is correlated with the error term.
 - ‘Confounder:’ This means that there is another unmeasured variable (a confounder) that affects the IV which also affects the DV.
 - We haven’t included this other confounding variable in our model.
 - Consequently, calling our observed correlation causal could be ‘spurious’ – it’s really a different variable, which we haven’t measured, that’s driving the results.
- To get our causal estimate, we need to create exogeneity:
 - We need our IV to be unrelated to (uncorrelated with) the error term.
 - We can do this through random assignment

Formalizing our Model

- Y_i = dependent variable, outcome variable, the thing we want to predict
- X_i = independent variable, explanatory variable, the thing that predicts the DV
- e_i = the error term, the part of the DV our IV doesn't explain, everything NOT in our model (usually a lot!). Note: the error term is not directly observable.

Core Statistical Model

REMEMBER THIS

Our core statistical model is

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

1. β_1 , the slope, indicates how much change in Y (the dependent variable) is expected if X (the independent variable) increases by one unit.
2. β_0 , the intercept, indicates where the regression line crosses the Y -axis. It is the value of Y when X is zero.
3. β_1 is almost always more interesting than β_0 .

β_1 - slope coefficient, relationship between X & Y
 β_0 - constant; value of Y when X is zero (intercept)

Our Core Model

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

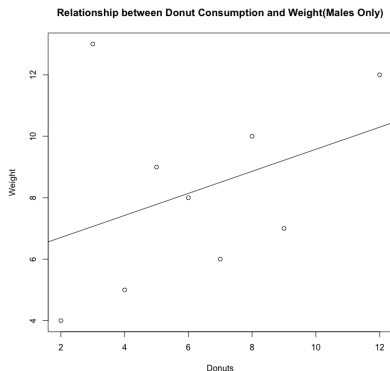
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Our Core Model

$$WeightGain_i = \beta_0 + \beta_1 DonutConsumption_i + \epsilon_i$$

- Simple Interpretation: Weight gain is a product of donut consumption plus other factors

The Core Model



$$WeightGain_i = \beta_0 + \beta_1 DonutConsumption_i + \epsilon_i$$

This regression line corresponds to the core model

Some Key Terms

- Causation
- Correlation
- Random Assignment
- Experiment
- Observational
- Dependent Variable
- Independent Variable
- Spurious
- Endogeneity
- Exogeneity
- Specification
- The Core Model
- Error Term

Any Questions?

Questions on causality? the core model?

